

Stefan

P1: Stefan

WCAG guidelines	Tools / Assistive technologies	What will the tool provide	Relevant automated checker tools	Does an API exist for this tool? (axecore, Pa11y but more?)	Link to ground truth test suite	If the test suite doesn't exist, link to manually written test suites
2.2.2 Pause, Stop, Hide	Reader view Pop up blockers Reduced motion settings (OS)	Controls that pause, stop, or hide animations that move, blink, or scroll.	axe	axe-core	https://www.w3.org/WAI/standards-guidelines/act/rules/efbfc7/proposed/	
3.1.3 Unusual Words	Text to Speech Reading Assistants	Read aloud the text and also provides immediate access to definitions, synonyms, and pronunciation of words.	axe, WAVE	Axe-core, WAVE API, textstat		https://www.w3.org/TR/WCAG20-TECHS/G101.html#G101-examples *examples but not test suites
3.1.4 Abbreviations	Text to Speech	Can expand acronyms to full words if coded with that function	axe, WAVE	axe-core		https://www.w3.org/WAI/WCAG21/Techniques/html/H28 https://www.w3.org/WAI/WCAG21/Techniques/html/H40#examples *not test suites, but

						have examples of good HTML
1.4.2 Audio Control	Reader view Pop up blockers	Controls that allow for pausing or stopping audio/video that plays automatically	Axe	axe-core	https://www.w3.org/WAI/standards-guidelines/act/rules/4c31df/proposed/	
1.4.8 Visual Presentation	Text to Speech Color overlays Font modifiers Reader view	Ability to change fonts and spacing for easier reading. Text read outloud for easier understanding.	Text spacing booklet + selenium to count errors	No – would probably have to write some code to inject booklet and check for infractions		Need manual
1.4.12 Text Spacing	Text to Speech Font modifiers Reader view	Text to speech can make it easier to process the text on webpage. Font modifiers and reader view can allow for easier reading experience.	Axe, WAVE	Axe-core, WAVE API	https://www.w3.org/WAI/standards-guidelines/act/rules/78fd32/ https://www.w3.org/WAI/standards-guidelines/act/rules/9e45ec/ https://www.w3.org/WAI/standards-guidelines/act/rules/24afc2/	

Ade

P5: Ade (Limited Mobility)

WCAG guidelines	Tools / Assistive technologies	What will the tool provide	Relevant automated checker tools	Does an API exist for this tool? (axecore, Pa11y but more?)	Link to ground truth test suite	If the test suite doesn't exist, link to manually written test suites
2.1.1 Keyboard (Operable)	Keyboard, switch interface, voice control	Sequential navigation (tab/shift+tab) and activation without a mouse.	Axe-core, WAVE	Yes (axe-core, Pa11y, Google Lighthouse)	https://www.w3.org/WAI/WCAG22/Understanding/keyboard	
2.5.3 Label in Name	Speech Recognition Software (Apple Voice Control)	Allows Ade to activate controls by speaking their visible text label.	Axe-core, Google Lighthouse, IBM Equal Access	Yes (axe-core rule: label-content-name-mismatch)	https://www.w3.org/WAI/sstandards-guidelines/act/rules/2ee8b8/proposed/	
2.5.5 Target Size (Enhanced)	Joystick, Adaptive Mouse, Touchscreen (Tablet)	Requires larger hit areas (44x44px+) to compensate for limited motor precision/tremors.	Axe-core, Android Accessibility Scanner	Yes (axe-core rule: target-size)	https://www.w3.org/WAI/WCAG21/Understanding/target-size.html https://www.w3.org/WAI/WCAG21/Techniques/css/C44	
2.2.1 Timing Adjustable	Keyboard	Mechanisms to turn off, adjust, or extend time limits before a session expires.	Axe-core (Checks for <meta http-equiv="refresh"> only).	Yes (for meta-refresh checks only)	https://www.w3.org/WAI/sstandards-guidelines/act/rules/bisz58/	
2.4.1 Bypass Blocks	Keyboard (tab)	A link to bypass repetitive	Axe-core, WAVE, Lighthouse	Yes (axe-core rule: skip-link)	https://www.w3.org/WAI/sstandards-guid	

		navigation menus.			elines/act/rules/cf77f2/proposed/	
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lan

P5: Ian (Autism)

WCAG guidelines	Tools / Assistive technologies	What will the tool provide	Relevant automated checker tools	Does an API exist for this tool? (axecore, Pa11y but more?)	Link to ground truth test suite	If the test suite doesn't exist, link to manually written test suites
2.2.2 Pause, Stop, Hide	Browser Developer Tools, Animation Blockers	Identifies moving content and checks for pause mechanisms.	Axe-core	Yes (axe-core rule: blink, marquee)	https://www.w3.org/WAI/standards-guidelines/act/rules/efbfc7/proposed/	
3.3.1 Error Identification	Keyboard/Mouse	Verifies if error messages are clearly described in text and linked to the correct field.	Axe-core, WAVE	Yes (axe-core rule: aria-valid-attr-value, form-field-multiple-labels)	https://www.w3.org/WAI/standards-guidelines/act/rules/36b590/proposed/	
1.3.5 Identify Input Purpose	Browser Autofill, Password Managers	Ensures fields have autocomplete attributes so Ian doesn't have to type/remember data.	Axe-core	Yes (axe-core rule: autocomplete-valid)	https://www.w3.org/WAI/standards-guidelines/act/rules/73f2c2/	
2.5.3 Label in Name	Speech recognition	Ensures the visible label matches the code, so if Ian reads a button, his speech tool recognizes it.	Axe-core	Yes (axe-core rule: label-content-name-mismatch)	https://www.w3.org/WAI/standards-guidelines/act/rules/2ee8b8/proposed/	
3.1.3 Unusual Words	Readability Analyzers	Highlights complex words that might confuse literal thinkers.	Acrolinx, Siteimprove (Content quality checkers, not pure accessibility)	Yes (Acrolinx API, Siteimprove API).		https://digital.gov/guides/plain-language

			tools).			
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Elias

Elias- (low-vision, low-hearing, slight hand tremor, minor short-term memory loss)

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2.2.2 Pause, Stop, Hide	Reader view Pop up blockers Reduced motion settings (OS)	Controls that pause, stop, or hide animations that move, blink, or scroll.	axe	axe-core	https://www.w3.org/WAI/standards-guidelines/act/rules/efbfc7/proposed/	
1.3.5 Identify Input Purpose (Level AA)	Alternative keyboard and mouse (input)	Typing accurately and navigating using a standard mouse are highly difficult with a hand tremor, alternate keyboard/ mouse can also take more time, making autofill important.	Testparty, Axe	Axe-core, (catches missing autocomplete) Testparty (Catches missing autocomplete + semi checks semantic value)	https://www.w3.org/WAI/standards-guidelines/act/rules/73f2c2/	
1.4.6: Contrast (Enhanced) (Level AAA)	Screen magnification	Can enlarge text to make it easier for reading.	axe, Pa11y	Axe-core Pa11y	https://www.w3.org/WAI/standards-guidelines/act/rules/09o5cg/	
1.4.2 Audio	Reader	Controls	Axe	axe-core	https://www	

Control	view Pop up blockers	that allow for pausing or stopping audio/video that plays automatically			w.w3.org/WAI/standards-guidelines/act/rules/4c31df/proposed/	
1.4.12 Text Spacing	Text to Speech Font modifiers Reader view	Text to speech can make it easier to process the text on webpage. Font modifiers and reader view can allow for easier reading experience .	Axe, WAVE	Axe-core, WAVE API	https://www.w3.org/WAI/standards-guidelines/act/rules/78fd32/ https://www.w3.org/WAI/standards-guidelines/act/rules/9e45ec/ https://www.w3.org/WAI/standards-guidelines/act/rules/24afc2/	

Lakshmi

Tab 2

Step 1: Run Tools (Objective)

- └─ NVDA announces "graphic"
- └─ Flesch score: 45
- └─ 3 animations detected

Step 2: Feed to LLM with Persona Context (Interpretation)

- └─ "You are Stefan. Given this data, what issues affect you?"

Step 3: LLM Returns STRUCTURED Output (Testable)

```
└─ {  
    "label": "FAIL",  
    "wcag_criteria": ["2.2.2", "3.1.5"],  
    "severity": "critical"  
}
```

Step 4: Validate Against Ground Truth

- └─ W3C ACT test case says: Expected = FAIL, WCAG = 2.2.2
- └─ Our output: FAIL, WCAG = 2.2.2

Plan

Step 1: Run tools on this HTML

```
tool_data = {  
    "nvda_navigation": [  
        {"key": "Tab", "announcement": "button"} # No name announced!  
    ],  
    "semantic": {  
        "button_has_accessible_name": False  
    }  
}
```

Step 2: Send to LLM

prompt = f"""

You are Lakshmi (blind screen reader user).

Tool Output:

```
{json.dumps(tool_data)}
```

Evaluate this page. Respond with ONLY this JSON format:

```
{{  
    "label": "PASS" | "FAIL" | "PARTIAL",  
    "wcag_criteria": ["1.1.1", ...],  
    "reasoning": ,  
    "severity": "critical" | "serious" | "moderate" | "minor",  
}}
```

```
    "remediation_instructions":  
  }  
}
```

```
# Step 3: LLM responds
```

```
llm_output = {  
    "label": "FAIL",  
    "wcag_criteria": ["1.1.1", "4.1.2"],  
    "severity": "critical"  
}
```

```
# Step 4: Validate
```

```
assert llm_output["label"] == "FAIL" # Matches expected  
assert "1.1.1" in llm_output["wcag_criteria"] # Correct criterion
```